

WILDFIRE

Wildfires are becoming increasingly common around the world. Rising global temperatures are causing hotter and drier environments that create ideal conditions for these deadly and destructive fires. Wildfires pose an immediate risk to anything in their path, while the smoke from wildfires presents a far-reaching and long-lasting risk, particularly to children.

As wildfires grow in size and frequency, and because of the smoke's ability to travel long distances, it's important that parents and caregivers learn how to protect their families from the harmful effects of wildfire smoke.

How wildfire smoke affects children

As children are growing, their developing immune systems and lungs make them especially vulnerable to the harmful effects of wildfire smoke, which is more toxic than day-to-day air pollution. The fine particulate matter released by wildfires – known as PM2.5 – can penetrate deeply into children's lungs affecting both their immediate and longer-term health and well-being.

Infants

Infants have developing bodies, breathe more air and are closer to the ground. They are at increased risk of:

Low birth weight (linked to later-life respiratory problems and reduced height-for-age)

Birth defects

Childhood diabetes

Death

Young children

Immune systems, brains and lungs are still developing at this stage. Children are also moving more and spending time outdoors. This increases their risk of:

Asthma and respiratory diseases

Weaker immune systems

Lower brain function

Adolescents

Adolescents continue to develop mentally and physically and are also more active.

They are now at risk of:

Asthma and respiratory diseases

Metabolic disorders, e.g. obesity

High blood pressure (hypertension)

Adverse mental health effects

Signs and symptoms of wildfire smoke exposure in children

Children – young children in particular – who breathe in wildfire smoke may experience a wide range of symptoms. Physical symptoms include:

Chest pain and tightness

Trouble breathing

Wheezing

Coughing

Burning or irritation in the nose, throat and eyes

Dizziness

If your child is experiencing breathing problems, is unable to sleep, or is experiencing noticeable health problems, take them to a health facility immediately.

How to protect children from wildfire smoke

Wildfires safety tips: Two parents and their baby in Belize

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Being prepared can help your family stay safe and together in an emergency. If you live in an area at risk of wildfire, you should review your plan with your family once a

year, so that everyone is prepared.

Always pay attention to local weather and air quality news and public health advisories, including those related to wildfire smoke. Know the symptoms of inhaling wildfire smoke, so you can act immediately to prevent further health problems.

Before a wildfire

Talk to your family about wildfires. Explain to your child in an age-appropriate way what wildfires are and how you will do everything you can to keep them safe.

Have an emergency evacuation plan ready, including medical information and what your family should do if separated. Identify safe shelters nearby or in neighbouring areas and prepare your children by ensuring that they have your contact information and know what to do if separated. Check with your child's school and learn what the school's plan is in case of a wildfire.

Create an "emergency kit" that includes non-perishable foods, medicines, cash, a first aid kit, a fire extinguisher, N95/KN95 masks (in adult and child sizes), flashlight, radio, batteries and water for several days in case you need to leave your area or services are cut off. Include any other necessary items to meet the needs of your family, as well as copies of important documents like birth certificates, ID papers, house deeds etc.

If you live in an area vulnerable to wildfires, for example near vegetation or a forest, create a fire-resistant zone around the house that is free of leaves, debris or flammable materials for at least 10 metres, if possible.

Evacuate immediately if authorities advise you to do so or if your home is at risk.

During a wildfire

Evacuate immediately if authorities advise you to do so or if your home is at risk.

Physically relocate to reduce exposure to wildfire smoke if feasible.

Keep your children indoors with the doors and windows closed, if possible. If using air conditioning, run it with the fresh air intake closed if possible, to keep outdoor smoke from getting indoors. Use a portable air cleaner if available.

Limit physical activity and drink water to stay hydrated.

If your child must go outdoors, consider using a N95/KN95 mask for protection.

These masks are often not designed for children, but those aged 7 and older may be able to wear small or extra small adult-sized masks. Children should not wear a mask if it causes breathing difficulty or if they are not able to remove the mask on their own.

After a wildfire

Don't let your children do any clean-up work after a wildfire. Ash and debris should be removed before they return home.

If your child starts experiencing any of the symptoms above, take them to a health facility immediately. Pay attention to any physical symptoms or emotional reactions in the months following.

Children may not express their symptoms in the same way as adults and therefore need to be handled with more patience and support after experiencing a traumatic event.

> See tips on comforting your child after a wildfire

Things to avoid

To limit exposure from other sources of air pollution during and immediately after a wildfire event, avoid the following:

Smoking

Using gas, propane or wood-burning stoves or fireplaces, or candles

Using ozone-generating air cleaners

Using natural-gas- or gasoline-powered generators indoors

Using spray cans

Frying or broiling food

Using the vacuum

Burning waste

Even in the absence of wildfires, air pollution around the home can affect a child's health drastically.

Prevention and Preparedness

- Keep emergency contact numbers of the district fire service department (101) and local forest authorities handy.
- Immediately inform the fire service department and local forest authorities in case of an unattended or out-of-control fire.
- Never leave a fire unattended in the vicinity of forested areas. Completely extinguish the fire by dousing it with water and stirring the ashes until it is cold.
- Be careful when using and fueling lanterns, stoves, and heaters while camping. Make sure lighting and heating devices are cooled before refuelling. Avoid spilling flammable liquids, and store fuel away from appliances.
- Do not discard cigarettes, matches and smoking materials while passing through forested areas.
- Do not burn stubble, municipal waste, etc., next to a forest area.
- Do not burn dry waste in farms close to forest areas.
- People living in remote and forest fire-prone areas should know their evacuation route(s) and keep emergency supplies ready.
- People living in remote and forest fire-prone areas should practice evacuation drills regularly in collaboration with local forest authorities.
- People living in remote and forest fire-prone areas should determine safe meeting locations in advance and make family members aware of such safe locations.

Evacuation in case of forest fire

- Evacuate immediately to the safer places on instructions from local authorities.
- Take precautions for protection from flying sparks and ashes.
- Untie cattle to ensure their safety.

Before leaving your house

- Remove all combustibles that may catch forest fire, including firewood, yard waste, cooking gas cylinders, fuel cans, etc., from your yard and relocate them to safer places.
- Close all windows, vents, and doors to prevent draft.

- Fill large vessels with water to limit the spreading of fire. It can also be used for emergency dousing.
- Cut down bushes and hedges close to the house to isolate the house from fire.
- Clear all gully and roof spaces of leaves. Spray water to dampen the house roof, walls and any trees and foliage adjacent to the house.
- Close shutters, blinds or any non-combustible window coverings to reduce radiant heat.
- Contact local forest authorities to know about forest fire incidents in your area. Pay attention to warnings and alerts issued by local officials about forest fires.

If caught in a forest fire in an open area:

- Don't try to outrun the blaze. Instead, look for a static water body such as a pond or river to take refuge until the fire passes. Do not take refuge in a fast river or stream.
- If there is no water nearby, find a depressed, clear area with little vegetation, lie low to the ground and cover your body with wet clothes, a blanket, or soil until the fire passes.
- Breathe through a moist cloth, if possible, to avoid inhaling smoke.
- Take shelter in an area with little or no fuel to feed the fire.

If trapped at home in a forest fire:

- Stay calm. As the fire front approaches, retreat inside the house. It is probable that the fire will pass before the house burns down.
- If a closed door is hot to touch, do not open it, as there may be fire on the other side

If caught in a vehicle in forest fire:

- Stay in your vehicle. It is less dangerous than trying to outrun a forest fire on foot.
- Close windows and air vents.
- Drive slowly with headlights full on.
- Watch carefully for other vehicles and people on foot.
- Do not drive through heavy smoke. If you can't see what's ahead, don't risk danger.

In case you must stop:

- Park as far as possible from trees and thickets.
- Keep lights on and the engine off.
- Stay on the floor of your vehicle.
- Cover your body with anything nonflammable.
- Do not panic if smoke and sparks enter the vehicle as fuel tanks rarely explode.

Always Remember-

- Do not smoke near thick vegetation; do not leave a lighted cigarette/ bidi in dry vegetation areas.
- Do not leave any open fire in the forest area after use.
- Lightning in places of dry vegetation causes fires, create a buffer zone to manage such fires, if possible, else alert forest officials and seek help.
- Try to put the fire out by digging or circling around it with water; if that is not possible, call the Fire brigade.
- Move farm animals & movable goods to safer places.
- During fire, listen regularly to news for advance information & obey the instructions cum advice given by local authorities regarding moving to safe locations.
- Teach your family, friends, and others about the causes and harm of fire. Make people aware of forest fire safety.
- Do not be scared when a sudden fire occurs in the forest. Be calm and encourage others and the community to deal with it patiently.
- After adventure activities, ensure no light bonfire is left.
- One should not leave the burning wood sticks in or near the forest vegetation.
- Don't enter the forest during the fire.
- Discourage the Community from using the slash & burn method for cultivation. This also has severe health impacts as Co2 levels rise